Bundle Structures in Topology

Labix

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Abstract

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Contents

1 Fibrations and Cofibrations

1.1 Fibrations and The Homotopy Lifting Property

Definition 1.1.1 (The Homotopy Lifting Property) Let $p : E \to B$ be a map and let X be a space. We say that p has the homotopy lifting property with respect to X if for every homotopy $H : X \times I \to B$ and a lift $\widetilde{H(-,0)} : X \to E$ of $H(-,0)$, there exists a homotopy $\widetilde{H} : X \times I \to E$ such that the following diagram commutes:

$$\begin{tikzcd} X \times \{0\} \arrow[r, "\widetilde{H(-,0)}"] \arrow[d, hook, "\iota"'] & E \arrow[d, "p"] \\ X \times I \arrow[r, "H"'] \arrow[ur, dashed, "\exists \widetilde{H}"] & B \end{tikzcd}$$

Definition 1.1.2 (Fibrations) We say that a map $p : E \to B$ is a fibration if it has the homotopy lifting property with respect to all topological spaces X. We call B the base space and E the total space.

Definition 1.1.3 (The Hopf Fibration) Define the Hopf fibration $h : S^3 \to S^2$ as follows. Consider S^2 as the one point compactification of $\mathbb{C}$. Also consider $S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}$. Define the map h by

$$(z_1, z_2) \to \frac{z_2}{z_1}$$

Example 1.1.4 The Hopf fibration $h : S^3 \to S^2$ is a fibration. Moreover, the fibers of the Hopf fibration are circles S^1.

Proof We can rewrite the coordinates of S^3 by $r_j e^{i\theta_j}$. Then

$$h(r_1 e^{i\theta_1}, r_2 e^{i\theta_2}) = \frac{r_2}{r_1} e^{i(\theta_2 - \theta_1)}$$

Fix $re^{i\theta} \in S^2$. Then there exists a unique pair (r_1, r_2) that solves the simultaneous equation $rr_1 = r_2$ and $r_1^2 + r_2^2 = 1$. ■

1.2 Cofibrations and The Homotopy Extension Property

Definition 1.2.1 (The Homotopy Extension Property) Let $i : A \to X$ be a map and let Y be a space. Denote i_0 the inclusion map $A \times \{0\} \hookrightarrow A \times I$. We say that i has the homotopy extension property with respect to Y if for every homotopy $H : A \times I \to Y$ and every map $f : X \to Y$ such that

$$H \circ i_0 = f \circ i$$

there exists a homotopy $\widetilde{H} : X \times I \to Y$ such that the following diagram commute:

$$\begin{tikzcd} A \cong A \times \{0\} \arrow[r, "\iota_0"] \arrow[d, hook, "i"'] & A \times I \arrow[d, "i \times \mathrm{id}_I"] \arrow[ddr, bend left, "H"] & \\ X \cong X \times \{0\} \arrow[r, "\iota_0"'] \arrow[drr, bend right, "f"'] & X \times I \arrow[dr, dashed, "\exists \tilde{H}"] & \\ & & Y \end{tikzcd}$$

Definition 1.2.2 (Cofibrations) Let A, X be spaces. Let $i : A \to X$ be a map. We say that i is a cofibration if it has the homotopy extension property for all spaces Y.

Proposition 1.2.3 Let A, X be spaces. Let $i : A \to X$ be a cofibration. Then $i : A \to i(A)$ is a homeomorphism.

There is actually an easier way to write out cofibrations when (X, A) is a pair of spaces.

Lemma 1.2.4 Let (X, A) be a pair of spaces with A closed in X. Let $\iota : A \to X$ be the inclusion. Then ι is a cofibration if and only if for all spaces Y and maps $f : X \to Y$ and $H : A \times I \to Y$, there exists a map $\tilde{H} : X \times I \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} X \times \{0\} \cup A \times I \arrow[r, "f \cup H"] \arrow[d, "\iota"'] & Y \\ X \times I \arrow[ur, "\tilde{H}"'] & \end{tikzcd}$$

1.3 Basic Properties of Fibrations and Cofibrations

Proposition 1.3.1 Let $X_1, X_2, Y_{,1}, Y_2$ be spaces. Let $p_1 : X_1 \to Y_1$ and $p_2 : X_2 \to Y_2$ be maps. Then the following are true.

- If p_1 and p_2 are fibrations then $p_1 \times p_2 : X_1 \times X_2 \to Y_1 \times Y_2$ is a fibration.
- If p_1 and p_2 are cofibrations then $p_1 \coprod p_2 : X_1 \coprod X_2 \to Y_1 \coprod Y_2$ is a cofibration.

Proposition 1.3.2 Let X, Y, Z be spaces. Let $f : X \to Y$ be a map.

- Let f be a fibration. Consider the following lifting problem:

$$\begin{tikzcd} Z \times \{0\} \arrow[r, "g"] \arrow[d, hook, "i_0"'] & X \arrow[d, "f"] \\ Z \times I \arrow[r, "h"'] \arrow[ur, dashed] & Y \end{tikzcd}$$

  If h_0 and h_1 are both solutions to the lifting problem, then h_0 and h_1 are homotopic relative to $Z \times \{0\}$.
- Let f be a cofibration. Consider the following extension problem:

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & Z \times \{0\} \\ Y \arrow[r, "h"'] \arrow[ur, dashed] & Z \times I \arrow[u, hook, "\text{ev}_0"'] \end{tikzcd}$$

  If h_0 and h_1 are both solutions to the extension problem, then h_0 and h_1 are homotopic relative to Z.

1.4 Serre Fibrations

Definition 1.4.1 (Serre Fibration) We say that a map $p : E \to B$ is a Serre fibration if it has the homotopy lifting property with respect to all CW-complexes.

Lemma 1.4.2 Every (Hurewicz) fibration is a Serre fibration.

Proof This is true since Hurewicz fibrations satisfies the homotopy lifting property with respect to all topological spaces, including CW complexes. ■

2 Vector Bundles

2.1 Basic Definitions

Definition 2.1.1 (Vector Bundles)
Let K be a field. Let E, B be spaces with B connected. Let $p : E \to B$ be a continuous map. We say that p is a vector bundle of dimension $d \in \mathbb{N}$ over K if the following are true.

- $p : E \to B$ is surjective.
- Local Triviality: There exists a collection

$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \to U_k \times F) \mid k \in I\}$$

called local trivialization charts such that the following diagram commutes:

$$\begin{tikzcd} p^{-1}(U_k) \arrow[rr, "\phi_k"] \arrow[dr, "p"'] & & U_k \times K^d \arrow[dl, "\pi"] \\ & U_k & \end{tikzcd}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B).

The local trivialization means that locally at a neighbourhood, the vector bundle looks the same the open set times F^k. In particular, there is also a notion of trivial bundle which means that the bundle is globally just $B \times \mathbb{R}^r$.

Lemma 2.1.2
Let K be a field. Let $p : E \to B$ be a vector bundle of dimension $d \in \mathbb{N}$ over K. Then the local trivialization charts induces a homeomorhpsim

$$p^{-1}(b) \cong K^d$$

for all $b \in B$.

Definition 2.1.3 (Sections)
Let k be a field. Let $p : E \to B$ be a vector bundle over k. A section of p is a map $s : B \to E$ such that $p \circ s = \mathrm{id}_B$.

Lemma 2.1.4
Let k be a field. Let $p : E \to B$ be a vector bundle over k. Then the following are true.

- Let $s_1, s_2 : B \to E$ be sections of p. Then $s_1 + s_2$ is a section of p.
- Let $s : B \to E$ be a section of p. Let $\lambda \in k$. Then λs is a section of p.

Definition 2.1.5 (The Space of Sections)
Let k be a field. Let $p : E \to B$ be a vector bundle over k. Define the vector space of sections to be the set

$$s(E) = \{s : B \to E \mid s \text{ is a section of } p\}$$

together with addition and scalar multiplication defined pointwise.

Proposition 2.1.6 Let $p : E \to B$ be a vector bundle. Let s, s_1, s_2 be sections of E. Then $s_1 + s_2$ and λs are also vector bundles for any $\lambda \in \mathbb{R}$. Moreover, the set of all sections $s(E)$ is a vector space.

Definition 2.1.7 (Morphism of Vector Bundles) Let $p_1 : E_1 \to B_1$ and $p_2 : E_2 \to B_2$ be vector bundles. A morphism of these vector bundles is given by is a pair of continuous maps $f : E_1 \to E_2$ and $g : B_1 \to B_2$ such that the following diagram commutes

$$\begin{tikzcd} E_1 \arrow[r, "f"] \arrow[d, "p_1"] & E_2 \arrow[d, "p_2"] \\ B_1 \arrow[r, "g"] & B_2 \end{tikzcd}$$

If $B = B_1 = B_2$ then the diagram collapses:

$$\begin{tikzcd} E_1 \arrow[rr, "f"] \arrow[dr, "p_1"] & & E_2 \arrow[dl, "p_2"] \\ & B & \end{tikzcd}$$

Definition 2.1.8 (Isomorphism of Vector Bundles) A bundle homomorphism from E_1 to E_2 is an isomorphism if there exists an inverse bundle homomorphism from E_2 to E_1. In this case, we say that E_1 and E_2 are isomorphic.

2.2 The Cocycle Conditions

Given two charts (U_α, ϕ_α) and (U_β, ϕ_β) of a vector bundle,

$$\phi_\beta \circ \phi_\alpha^{-1} : (U_\alpha \cap U_\beta) \times F^k \to (U_\alpha \cap U_\beta) \times F^k$$

is a well defined function. In particular, by fixing a point in $U_\alpha \cap U_\beta$, we obtain a linear map.

Definition 2.2.1 (Transition Functions) Let $p : E \to B$ be an F-vector bundle of rank r. Let (U_α, ϕ_α) and (U_β, ϕ_β) be local trivialization. For each $x \in U_\alpha \cap U_\beta$, $\phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \to F^k$ is a linear map. Define $g_{U_\alpha U_\beta} : U_\alpha \cap U_\beta \to \text{GL}(n, F)$ by

$$x \mapsto \phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \to F^k$$

In other words, $g_{U_\alpha U_\beta}$ is such that

$$\phi_\beta \circ \phi_\alpha^{-1}(x, v) = (x, g_{U_\alpha U_\beta}(x)v)$$

For $x \in U_\alpha \cap U_\beta$ and $v \in F^k$.

Proposition 2.2.2 Let $p : E \to B$ be a K-vector bundle of rank r. The transition functions of the vector bundle satisfies the following.

- Cocycle condition: $g_{\alpha\beta} \circ g_{\beta\gamma} \circ g_{\gamma\alpha} = I_r$ on $U_\alpha \cap U_\beta \cap U_\gamma$
- $g_{\alpha\alpha} = I_r$ on U_α

2.3 Operations on Vector Bundles

Definition 2.3.1 (Whitney Sum) Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be two vector bundles. Define the direct sum of the vector bundles to be

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

together with the projection $p : E_1 \oplus E_2 \to B$ defined by $(v_1, v_2) \mapsto p_1(v) = p_2(v)$.

Lemma 2.3.2 The Whitney sum $E_1 \oplus E_2$ of two vector bundles is again a vector bundle.

Proposition 2.3.3 (Tensor Product Bundle) Let $p_1 : E_1 \to B$ and $p_2 : E_2 \to B$ be vector bundles. Define the tensor product bundle of it to be

$$E_1 \otimes E_2 = \{p_1^{-1}(x) \otimes p_2^{-1}(x) | x \in B\}$$

The construction $E_1 \otimes E_2$ is a vector bundle over B.

Theorem 2.3.4 (Pullback Bundle) Let $p : E \to Y$ be a vector bundle. Let $f : X \to Y$ be a continuous map. Then there exists E' and p' such that $p' : E' \to X$ is a vector bundle.

Theorem 2.3.5 (Dual Bundle) Let $p : E \to B$ be a K-vector bundle. Then the dual bundle $p^* : E^* \to B$ defined by

$$E_b^* = \mathrm{Hom}_K(E_b, K)$$

is a vector bundle over B.

3 The Topology of Fiber Bundles

3.1 Fiber Bundles

Fiber bundles serve as somewhat of a generalization of both vector bundles and covering spaces, while being a special case of a fibration. It therefore has the properties of a fibration.

Definition 3.1.1 (Fiber Bundles)
Let E, B, F be spaces with B connected, and $p : E \to B$ a continuous map. We say that p is a fiber bundle over F if the following are true.

- $p : E \to B$ is surjective
- Local Triviality: There exists a collection
$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \xrightarrow{\cong} U_k \times F) \mid k \in I\}$$
called local trivialization charts such that the following diagram commutes:
$$\begin{tikzcd} p^{-1}(U_k) \arrow[rr, "\phi_k"] \arrow[dr, "p"'] & & U_k \times F \arrow[dl, "\pi"] \\ & U_k & \end{tikzcd}$$
where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) or $F \hookrightarrow E \to B$.

Intuitively, we would like a fiber bundle to locally look like the product $B \times F$.

Lemma 3.1.2
Let K be a field. Let $p : E \to B$ be a fiber bundle with fiber F. Then the local trivialization charts induces a homeomorhpsim
$$p^{-1}(b) \cong F$$
for all $b \in B$.

Fiber bundles generalizes vector bundles in the sense that the fibers are no longer vector spaces but instead arbitrary spaces.

Lemma 3.1.3
The following are true regarding fiber bundles.

- Every vector bundle is a fiber bundle.
- Every covering space is a fiber bundle.

Proof Indeed if $p : E \to B$ is a vector bundle, then each fiber $p^{-1}(b)$ is an n-dimensional vector spaces over a field F. Moreover, by definition the local triviality condition is also satisfied.

Finally, for any $U \subseteq X$, recall that
$$p^{-1}(U) = \coprod_{i \in I} V_i$$
where each $V_i \cong U$. It is clear by definition that $|p^{-1}(x)||I|$ for any $x \in X$. By giving I the discrete topology, we obtain a homeomorphism $p^{-1}(x) \cong I$. The homeomorphism $p^{-1}(U) = \coprod_{i \in I} V_i$ translates to
$$p^{-1}(U) = \coprod_{i \in I} V_i \cong \coprod_{i \in I} U \cong U \times I$$
defined by $\tilde{x} \in V_i \mapsto (p(\tilde{x}) = x, i)$. It is thus clear that the local triviality condition is satisfied. ■

Example 3.1.4
The following maps are fiber bundles.

- Let $f_n : S^1 \to S^1$ be the map defined by $f_n(z) = z^n$. Then f_n is a fiber bundle with fiber the

nth roots of unity.

- Let $f : \mathbb{R} \to S^1$ be the map defined by $f(t) = e^{2\pi i t}$. Then f is a fiber bundle with fiber $\mathbb{Z}$.
- Let $p : S^n \to \mathbb{RP}^n$ be the projection map. Then p is a fiber bundle with fiber as a two point set with the discrete topology.

Proof

Indeed, all of these are covering spaces and hence fiber bundles by the above lemma. ∎

Example 3.1.5

Let $M = \frac{[-1,1]\times[-1,1]}{(t,-1)\sim(1-t,1)}$ be the Mobius strip. Then the projection map $p : M \to C$ defined by $p(t,s) = s$ is a fiber bundle with fiber $[-1,1]$.

Proposition 3.1.6 Every fiber bundle is a Serre fibration.

We can provide a partial converse for the fact that every fiber bundle is a Serre fibration.

Proposition 3.1.7 Let $p : E \to B$ be a fiber bundle. If B is paracompact, then p is a (Hurewicz) fibration.

We there fore have inclusions

$$\substack{\text{Fiber}\\\text{Bundles}} \subset \substack{\text{Serre}\\\text{Fibrations}} \subset \substack{\text{(Hurewicz)}\\\text{Fibrations}}$$

Definition 3.1.8 (Map of Fiber Bundles)

Let (F_1, E_1, B_1) and (F_2, E_2, B_2) be fiber bundles. A map of fiber bundles is a pair of basepoint preserving continuous maps $(\tilde{f} : E_1 \to E_2, f : B_1 \to B_2)$ such that the following diagram commutes:

$$\begin{tikzcd}
E_1 \arrow[r, "\tilde{f}"] \arrow[d, "p_1"'] & E_2 \arrow[d, "p_2"] \\
B_1 \arrow[r, "f"'] & B_2
\end{tikzcd}$$

A map of fiber bundles $(\tilde{f}, f)$ is said to be an isomorphism if there is a map $(\tilde{g} : E_2 \to E_1, g : B_2 \to B_1)$ such that $\tilde{g}$ is the inverse of $\tilde{f}$ and g is the inverse of f.

Such a map of fiber bundles determine a continuous of the fibers $F_1 \cong p_1^{-1}(b_1) \to p_2^{-1}(b_2) \cong F_2$. Moreover, such a morphism preserves fibers. Indeed, If $p_1^{-1}(b)$ is a fiber of B, then using the commutativity of the diagram we have that

$$p_2(\overline{f}(p_1^{-1}(b))) = f(p_1(p_1^{-1}(b))) = f(b)$$

which implies that

$$p_2^{-1}(f(b)) = \overline{f}(p^{-1}(b))$$

or in other words, the fiber at $f(b)$ is the same as the fiber at b applied with $\overline{f}$.

Definition 3.1.9 (Isomorphic Fiber Bundles)

Let $p_1 : E_1 \to B_1$ and $p_2 : E_2 \to B_2$ be two fiber bundles. Let $(\widetilde{f}, f) : p_1 \to p_2$ be a map of fiber bundles. We say that $(\widetilde{f}, f)$ is an isomorphism if there exists a map $(\widetilde{g}, g) : p_2 \to p_1$ of fiber bundles such that $\widetilde{f}$ is the inverse of $\widetilde{g}$ and f is the inverse of g. In this case, we say that p_1 and p_2 are isomorphic.

There are two important special cases of fiber bundles that will appear time and time again.

Definition 3.1.10 (Trivial Bundles)

We say that a fiber bundle (F, E, B) is trivial if (F, E, B) is isomorphic to the trivial fibration $B \times F \to B$.

Definition 3.1.11 (The Pullback Bundle) Let $p : E \to B$ be a fiber bundle with fiber F. Let $f : B' \to B$ be a continuous function. Define the pullback of p by f to be the space

$$f^*(E) = \{(b', e) \in B' \times E \mid p(e) = f(b')\}$$

Theorem 3.1.12 Let $p : E \to B$ be a fiber bundle. Suppose that $f, g : X \to B$ are homotopic maps. Then the pull back bundles

$$f^*(E) \cong g^*(E)$$

are equivalent.

3.2 Sections of a Bundle

Definition 3.2.1 (Sections) Let (F, E, B) be a fiber bundle. A section on the fiber bundle is a map $s : B \to E$ such that

$$p \circ s = \text{id}_B$$

Definition 3.2.2 (Local Sections) Let (F, E, B) be a fiber bundle. Let $U \subset B$ be an open set. A local section of the fiber bundle on U is a map $s : U \to B$ such that

$$p \circ s = \text{id}_U$$

3.3 Sphere Bundles

We now consider a special type of fibrations where the fibers are given by S^1. We define a class of fiber bundles of the form $S^1 \to S^{2n+1} \to \mathbb{CP}^n$. When $n = 1$, this is the Hopf fibration.

Definition 3.3.1 (Sphere Bundles) A sphere bundle is a fiber bundle $p : E \to B$ for which its fibers are the n-sphere S^n.

Example 3.3.2
Let $n \in \mathbb{N}$. Consider S^{2n+1} lying inside $\mathbb{C}^{n+1}$. The projection map $\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{CP}^n$ induces a fiber bundle

$$S^1 \hookrightarrow S^{2n+1} \to \mathbb{CP}^n$$

Proof
Clearly the projection map $p : S^{2n+1} \to \mathbb{CP}^n$ is surjective. Consider the open cover $\{U_i \mid 0 \leq i \leq n\}$ where $U_i = \{[z_0 : \cdots : z_n] \mid z_i \neq 0\}$. Define the local trivialization charts $\phi_i : p^{-1}(U_i) \to U_i \times S^1$ by $(z_0, \dots, z_n) \to ([z_0 : \cdots : z_n], z_i/|z_i|)$. It is continuous since it is the product of the composition map and the projection map to the sphere S^1. Its inverse is the continuous map $([z_0, \dots, z_n], w) = w|z_i|z_i^{-1}(z_0, \dots, z_n)$ since it is induced by the universal property of quotients. It is clear that the required diagram commutes, that is $p = \text{proj} \circ \phi_i$ where $\text{proj} : U_i \times S^1 \to U_i$ is the projection map to the first coordinate. ■

Example 3.3.3
For each $n \in \mathbb{N}$, we have a map of fiber bundles:

$$\begin{tikzcd} S^{2n+1} \arrow[r, hook] \arrow[d] & S^{2n+3} \arrow[d] \\ \mathbb{CP}^n \arrow[r, hook] & \mathbb{CP}^{n+1} \end{tikzcd}$$

whose horizontal maps are induced by the inclusion $\mathbb{C}^n \hookrightarrow \mathbb{C}^{n+1}$ into the first n coordinates. Taking colimit of the diagram, we have that

$$S^1 \hookrightarrow S^\infty \to \mathbb{CP}^\infty$$

is a fiber bundle.

Proof

Since the local triviality condition is a local condition, for each $x \in \mathbb{CP}^\infty$, x lives in $\mathbb{CP}^n$ for some $n \in \mathbb{N}$ and since $S^1 \hookrightarrow S^{2n+1} \to \mathbb{CP}^n$ is a fiber bundle, there exists a chart for which the local triviality condition is satisfied. ■

Example 3.3.4 (Hopf Fibration / Hopf Bundle)

Consider the above example with $n = 1$. Restricting the total space to S^3, the fiber bundle $p : S^3 \to S^2$ with fiber S^1 is called the Hopf fibration / Hopf bundle.